AFS–0546

Sub. Code
7MMA1C1

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Mathematics

ALGEBRA – I

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is meant by a Normal subgroup?

2. Define an isomorphic of groups.

3. Define an internal direct product of groups.

4. State the first part of Sylow's theorem.

5. Give an example of a finite non-commutative ring.

6. State the Pigeonhold principle.

7. When will you say a ring is maximal ideal?

8. Define two-sided ideal of a ring.

9. State unique factorization theorem.

10. Find the greatest common divisor of the polynomials x^2+1 and x^6+x^3+x+1 in $Q[x]$.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) State and prove the Euler's theorem. Also state the Fermat theorem.

Or

(b) State and prove the Cauchy's theorem for Abelian group.

12. (a) If p is a prime number and $p / O(G)$, then prove that G has an element of order p.

Or

(b) State and prove second part of Sylow's theorem.

13. (a) Define an associative ring with an example. Let $a \in R$ and R has a unit element 1. Prove that :

(i) $(-1)a = -a$;

(ii) $(-1)(-1) = 1$.

Or

(b) Prove that the homomorphism φ of R into R' is an isomorphism if and only if $I(\varphi) = (0)$.

14. (a) If F is a field, then prove that its only ideals are (0) and F itself.

Or

(b) Prove that the mapping $\varphi : D \to F$ defined by $\varphi(a) = [a, 1]$ is an isomorphism of D into F.

15. (a) If p is a prime number of the form $4n+1$, then prove that $p = a^2 + b^2$ for some integers a, b.

Or

(b) State and prove the Einstein criterion theorem.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. State and prove the Cayley's theorem.

17. Let G be a group and suppose that G is the internal direct product of $N_1, N_2, \ldots N_n$. Let $T = N_1 \times N_2 \times \ldots \times N_n$. Prove that G and T are isomorphic.

18. (a) Define the following terms :

(i) Integral domain;

(ii) Division ring ;

(iii) Zero-divisor;

(iv) Homomorphism of rings;

(v) Isomorphic rings.

(b) If p is a prime number then prove that J_p, the ring of integers $\bmod p$, is a field.

19. Show that every integral domain can be imbedded in a field.

20. (a) With the usual notations, prove that $J[i]$ is a Euclidean ring.

(b) If R is a unique factorization domain, then prove that $R[x]$ is also an unique factorization domain.

AFS–0547

Sub. Code
7MMA1C2

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Mathematics

ANALYSIS – I

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Prove that every neighborhood is an open set.

2. Define the cantor set. Give an example.

3. When will you say a metric space is complete?

4. If $p > 0$, then prove that $\lim_{n\to\infty} \frac{1}{n^p} = 0$.

5. Define a power series. Give an example.

6. What is meant by a rearrangement of Σa_n ?

7. Define the uniformly continuous function.

8. Define the two kinds of discontinuities of a function.

9. Let f be defined on $[a, b]$. If f is differentiable at a point $x \in [a, b]$, then prove that f is continuous at x.

10. State the L' Hospital's rule.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Prove that a set E is open if and only if its complement is closed.

Or

(b) Show that compact subsets of metric spaces are closed.

12. (a) Suppose $\{s_n\}$ is monotonic. Prove that $\{s_n\}$ converges if and only if it is bounded.

Or

(b) Define e. Prove that e is irrational.

13. (a) Derive the partial summation formula.

Or

(b) If Σa_n is a series of complex numbers which converges absolutely, then prove that every rearrangement of Σa_n converges, and they all converge to the same sum.

14. (a) Let f be a continuous mapping of a compact metric space X into a metric space Y. Prove that f is uniformly continuous on X.

Or

(b) Show that a continuous image of a connected subset of a metric space X is connected.

15. (a) State and prove the chain rule for differentiation.

Or

(b) Examine for differentiability the function f given by $f(x) = \begin{cases} x \sin\frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Also verify that f is continuous at 0.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) With the usual notations, prove that every k-cell is compact.

(b) State and prove the Weierstrass theorem.

17. State and prove the Root and Ratio tests.

18. State and prove the Mertens theorem.

19. Prove that a mapping f of a metric space X into a metric space Y is continuous on X if and only if $f^{-1}(V)$ is open in X for every open set V in Y.

20. State and prove Taylor's theorem.

AFS–0548

Sub. Code
7MMA1C3

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Mathematics

DIFFERENTIAL GEOMETRY

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define arc length.

2. Write down the formula for the centre of spherical curvature.

3. What is meant by the pitch of the helicoid?

4. Define the anchor ring.

5. Define geodesic.

6. Give the Christoffel symbols of the second kind.

7. Write short notes on the geodesic curvature.

8. Define a pseudo – sphere.

9. State the Meusnier's theorem.

10. Define the following terms:

    (a) Edge of regression;

    (b) Osculating developable of the curve.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Find the curvature and Torsion of any curve $\bar{r} = \bar{r}(s)$.

Or

(b) Determine the intrinsic equations of the curve $\bar{r} = \left[ae^u \cos u,\ ae^u \sin u,\ be^u\right]$.

12. (a) Determine the coefficients of the direction which makes an angle $\pi/2$ with the direction whose coefficients are (l, m).

Or

(b) Show that on the right helicoid, the family of curves orthogonal to the curves $u \cos v = \text{constant}$ is the family $\left(u^2 + v^2\right)\sin^2 v = \text{constant}$.

13. (a) Prove that on a general surface, a necessary and sufficient condition that the curve $v = c$ be a geodesic is $EE_2 + FE_1 - 2EF_1 = 0$ where $v = c$, for all values of u.

Or

(b) Enumerate the normal property of geodesics.

14. (a) Prove that the components λ, μ of the geodesic curvature vector are given by the following formula with s as parameter.

$$\lambda = \frac{1}{H^2}\frac{U}{v'}\frac{\partial T}{\partial v'} = -\frac{1}{H^2}\frac{V}{u'}\frac{\partial T}{\partial v'};$$
$$\mu = \frac{1}{H^2}\frac{V}{u'}\frac{\partial T}{\partial u'} = -\frac{1}{H^2}\frac{U}{v'}\frac{\partial T}{\partial u'}.$$

Or

(b) Show that the surface generated by the tangents to any space curve is a surface of constant zero curvature.

15. (a) State and prove the Euler's theorem.

Or

(b) Prove that the edge of regression of the rectifying developable has equation $\overline{R} = \overline{r} + k\dfrac{\left(\tau\overline{t} + k\overline{b}\right)}{k'\tau - k\tau'}$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Obtain the curvature and torsion of the curve of intersection of the two quadric surfaces.

$ax^2 + by^2 + cz^2 = 1$, $a'x^2 + b'y^2 + c'z^2 = 1$.

17. Find a surface of revolution which is isometric with a region of the right helicoid.

18. Find the geodesics on a surface of revolution.

19. State and prove the Gauss – Bonnet theorem.

20. State and prove the Monge's theorem.

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AFS–0549

Sub. Code
7MMA1C4

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Mathematics

ORDINARY DIFFERENTIAL EQUATIONS

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Find all real valued solutions of the equation $y''-y=0$.

2. The functions $\phi_1(x)=\cos x$ and $\phi_2(x)=\sin x$ exist for $-\infty<x<\infty$. Determine whether they are linearly dependent or independent.

3. Find two linearly independent solutions of $y''+\frac{1}{4x^2}y=0, (x>0)$.

4. Show that $P_n(-x)=(-1)^n P_n(x)$.

5. Compute the indicial polynomial and its roots for the differential equation $x^2y''+xy'+(x^2-\frac{1}{4})y=0$.

6. Find the singular point of the equation $x^2y''-y'-\frac{3}{4}y=0$. Is this singular point regular? Justify your answer.

7. Solve $y'=3y^{2/3}$.

8. Show that the function f given by $f(x,y)=y^{1/2}$ does not satisfy a lipschitz condition on $R:|x|\le 1, 0\le y\le 1$.

9. Let $f(x,y)=\frac{\cos y}{1-x^2}, (|x|<1)$. Show that f satisfies a Lipschitz condition on every strip $S_a:|x|\le a$, where $0<a<1$.

10. State the local existence theorem for the initial value problem $y'=f(x,y), y(x_0)=y_0$ on Z.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) If ϕ_1,ϕ_2 are two solutions of $L(y)=0$ on an interval I. Containing a point x_0, then prove that

$$W(\phi_1,\phi_2)(x)=e^{-a,(x-x_0)}W(\phi_1,\phi_2)(x_0).$$

Or

(b) Let $\alpha_1, \alpha_2, \ldots\ldots \alpha_n$ be any n constants, and let x_0 be any real number. Prove that there exists a solution ϕ of $L(y) = 0$ on $-\infty < x < \infty$ satisfying $\phi(x_0) = \alpha_1, \phi'(x_0) = \alpha_2, \ldots\ldots \phi^{(n-1)}(x_0) = \alpha_n$.

12. (a) One solution of $x^2 y'' - 2y = 0$ on $0 < x < \infty$ is $\phi_1(x) = x^2$. Find all solutions of $x^2 y'' - 2y = 2x - 1$ on $0 < x < \infty$.

Or

(b) Show that $\int_{-1}^{1} P_n(x) P_m(x) dx = 0$, when $n \neq M$.

13. (a) Find all solutions of the equation $x^2 y'' + xy' - 4\pi y = x$ for $x > 0$.

Or

(b) Show that $x^{1/2} J_{1/2}(x) = \frac{\sqrt{2}}{\Gamma(1/2)} Sin\, x$ and $x^{1/2} J_{-1/2}(x) = \frac{\sqrt{2}}{\Gamma(1/2)} \cos x$.

14. (a) Consider the initial value problem $y' = 3y + 1, y(0) = 2.$

(i) Show that all the successive approximations $\phi_0, \phi_1, \ldots\ldots$ exist for all real x.

(ii) Compute the first four approximations $\phi_0, \phi_1, \phi_2, \phi_3$ to the solution.

Or

(b) Verify whether the equation $2xydx + (x^2 + 3y^2)dy = 0$ is exact or not, if exact solve.

15. (a) Let f be a continuous and satisfy a Lipschitz condition on R. If ϕ and ψ are two solutions of $y' = f(x, y), y(x_0) = y_0,$ on an interval I containing $x_{o,}$, then prove that $\phi(x) = \psi(x)$ for all x in I.

Or

(b) Consider the system $y_1' = 3y_1 + xy_3, y_2' = y_2 + x^3 y_3, y_3' = 2xy_1 - y_2 + e^x y_3.$ Show that every initial value problem for this system has a unique solution which exists for all real x.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Let b be continuous on an interval I. Prove that

(a) every solution ψ of $L(y) = b(x)$ on I can be written as $\psi = \psi_p + c_1\phi_1 + c_2\phi_2.,$, where ψ_p is a particular solution ϕ_1, ϕ_2 are two linearly independent solutions of $L(y) = 0,$ and c_1, c_2, are constants.

(b) a particular solution ψ_p is

$$\psi_p(x) = \int_{x_0}^{x} \frac{[\phi_1(t)\phi_2(x) - \phi_1(x)\phi_2(t)]}{W_{(\phi_1,\ \phi_2)^{(t)}}} b(t)dt.$$

(c) every such ψ is a solution of $L(y) = b(x)$.

17. Let ϕ_1 be a solution of $L(y) = 0$ on an interval I, and suppose $\phi_1(x) \neq 0$ on I. If $v_2.....v_n$ is any basis on I for the solutions of the linear equation.

$\phi_1 v^{(n-1)} + ... + [n\phi_1^{(n-1)} + a_1^{(n-1)}\phi_1^{(n-2)} + ... + a_{n-1}\phi_1]v = 0.$ of order $n-1,$ and if $v_k = u_k^{'}, (k = 2,.....n)$ then prove that $\phi_1, u_2\phi_1,.....u_n\phi_1$ is a basis for the solutions of $L(y) = 0$ on I.

18. Derive Bessel's function of first kind of order $\alpha, J_\alpha(x)$.

19. Let M, N be two real valued functions which have continuous first partial derivatives on some rectangle $R: |x - x_0| \le a, |y - y_0| \le b.$ prove that the equation $M(x,y) + N(x,y)y' = 0$ is exact in R if, and only if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ in R.

20. Let f be a continuous vector–valued function defined on $S: |x - x_0| \le a, |y| < \infty, (a >),$ and satisfy these a Lipschitz condition, prove that the successive approximations $\{\phi_k\}$ for the problem $y' = f(x, y), y(x_0) = y_0$ $(|y_0| < \infty),$ exist on $|x - x_0| \le a,$ and converge there to be a solution ϕ of this problem.

AFS–0550

Sub. Code
7MMA1E1

M.Sc. DEGREE EXAMINATION, APRIL 2018

First Semester

Mathematics

Elective – NUMBER THEORY

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. When will you say a number is composite? Give an example.

2. Define a reduced fraction.

3. State the Euler totient function ϕ (n).

4. When will you say the functions are divisor functions?

5. Define the Riemann zeta function.

6. State the legendre's identity.

7. What are the solutions of the congruence. $x^2 \equiv 1 \pmod 8$?

8. State the little fermat theorem.

9. Write down the legendre's symbol (n/p).

10. Determine whether –104 is a quadratic residue or nonresidue of the prime 997.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Show that there are infinitely many prime numbers.

Or

(b) State and prove the division algorithm.

12. (a) State and prove the mobius inversion formula.

Or

(b) Derive the selberg identity.

13. (a) If $x \geq 1$, then prove that $\sum_{n>x} \frac{1}{n^s} = o(x^{1-s})$ if $s > 1$.

Or

(b) Show that the set of lattice points visible from the origin has density $6/\pi^2$.

14. (a) Solve the congruence $25x \equiv 15 (\text{mod } 120)$.

Or

(b) State and prove the chinese remainder theorem.

15. (a) Show that legendre's symbol (n/p) is a completely multiplicative function of n.

Or

(b) State and prove reciprocity law for Jacobi symbols.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. (a) State and prove the Euclidean algorithm.

(b) Show that the infinite series $\sum_{n=1}^{\infty} 1/p_n$ diverges.

17. Define liouville's function $\lambda(n)$. for every $n \geq 1$,

Prove that $\sum_{d/n} \lambda(d) \begin{cases} 1 & \text{if } n \\ 0 & \text{otherwise} \end{cases}$ is a square,

Also prove $\lambda^{-1}(n) = |\mu(n)|$ for all n.

18. state and prove the Euler's summation formula.

19. (a) State and prove the wilson's theorem.

(b) State and prove the wolstenholme's theorem.

20. Prove that the diophantine equation $y^2 = x^3 + k$ has no solutions if k has the form $k = (4n-1)^3 - 4m^2$, where m and n are integers such that no prime $p \equiv -1 (\text{mod}\, 4)$ divides m.

AFS–0553

Sub. Code
7MMA2C1

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Mathematics

ALGEBRA – II

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define a vector space over a field F. give an example.

2. When will you say two vector spaces are isomorphic?

3. What is meant by an inner product space?

4. Define the following terms:

   (a) Dual space;

   (b) Linear functional.

5. Define an algebraic extension of F.

6. Is any finite extension of a field of characteristic 0 is a simple extension? Justify your answer.

7. Define the Galois group of $p(x)$.

8. What is meant by a normal extension of F.

9. Define a characteristic root.

10. When will you say a linear transformation is Unitary?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) If V is a vector space over F, then prove that

(i) $\alpha 0 = 0$ for $\alpha \in F$;

(ii) $0v = 0$ for $u \in V$;

(iii) $(-\alpha)v = -(\alpha v)$ for $\alpha \in F$, $v \in V$;

(iv) If $v \neq 0$, then $\alpha v = 0$ implies that $\alpha = 0$.

Or

(b) If $v_1, v_2, \ldots v_n$ is a basis of V over F and if $w_1, w_2, \ldots w_m$ in V are linearly independent over F, then prove that $m \leq n$.

12. (a) Prove that $A(W)$ is a subspace of $\hat{V}$.

Or

(b) State and prove the Schwarz inequality.

13. (a) Let $f(x) \in F[x]$ be of degree $n \geq 1$. Prove that there is an extension E of F of degree at most $n!$ in which $f(x)$ has n roots.

Or

(b) If $f(x) \in F[x]$ is irreducible, then prove that

(i) If the characteristic of F is 0, $f(x)$ has no multiple roots.

(ii) If the characteristic of F is $p \neq 0$, $f(x)$ has a multiple root only if it is of the form $f(x) = g(x^p)$.

14. (a) Show that the fixed field of G is a subfield of k.

Or

(b) Prove that k is a normal extension of F if and only if k is the splitting field of some polynomial over F.

15. (a) If $\lambda \in F$ is a characteristic root of $T \in A(V)$, then prove that for any polynomial $q(x) \in F[x]$, $q(\lambda)$ is a characteristic root of $q(T)$.

Or

(b) (i) If $T \in A(V)$ is Hermitian, then prove that all its characteristic roots are real.

(ii) If T is Hermitian and $vT^k = 0$ for $k \geq 1$ then prove that $vT = 0$.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. If V is finite –dimensional and it W is subspace of V, then prove that W is finite-dimensional, $\dim W \leq \dim V$ and $\dim {}^{V}\!/_{W} = \dim V - \dim W$.

17. Discuss the Gram-Schmidt orthogonalization process.

18. If L is a finite extension of k and if k is finite extension of F, then prove that L is a finite extension of F. Also prove $[L:F] = [L:K][K:F]$.

19. (a) If k is a finite extension of F, then prove that $G(K,F)$ is a finite group and its order, $0(G(K,F))$ satisfies $0(G(K,F)) \leq [K:F]$.

(b) If $P(x) = x^n - 1$ prove that the Galoris group of $P(x)$ over the field of rational numbers is abelian.

20. If V is n-dimensional over F and if $T \in A(V)$ has the matrix $m_1(T)$ in the basis $v_1, v_2, \ldots v_n$ and the matrix $m_2(T)$ in the basis $w_1, w_2, \ldots w_n$ of V over F, then prove that there is an element $C \in F_n$ such that $m_2(T) = Cm_1(T)C^{-1}$.

AFS–0554

Sub. Code
7MMA2C2

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Mathematics

ANALYSIS – II

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define a partition P of $[a, b]$.

2. Define the unit step function.

3. Show by an example that $\lim_{n\to\infty}\lim_{m\to\infty} S_{m,n} \neq \lim_{m\to\infty}\lim_{n\to\infty} S_{m,n}$.

4. Define a uniformly bounded sequence on a set E.

5. Show that $E(Z+W) = E(Z)E(W)\ \forall Z, W \in \mathbb{C}$

6. Prove that $|E(ix)| = 1\ \forall x \in R$.

7. If A and B are two sets in M with ACB, then prove that $mA \leq mB$.

8. If $m * E = 0$, then prove that E is measurable.

9. Define a Simple function.

10. When we say that a sequence $< fn >$ of measurable functions converge to f in measure?

Part B (5 × 5 = 25)

Answer **all** the questions, choosing either (a) or (b).

11. (a) If P* is a refinement of P, then prove that $L(p, f, \alpha) \leq L(P^*, f, d)$.

Or

(b) State and prove integration by parts theorem.

12. (a) Let α be monotonically increasing on $[a,b]$. Suppose $f_n \in \mathcal{R}(\alpha)$ on $[a,b]$ for $n = 1,2,3,.....$ and suppose $f_n \to f$ uniformly on $[a,b]$. Then prove that $f \in \mathcal{R}(\alpha)$ on $[a,b]$, and $\int_a^b f d\alpha = \lim_{n\to\infty} \int_a^b f_n d\alpha$.

Or

(b) If K is a compact metric space, if $f_n \in \mathcal{C}(K)$ for n=1,2,3.... and if $\{f_n\}$ converges uniformly on K, Then prove that $\{f_n\}$ is equi continuous on K.

13. (a) Suppose $\sum C_n$ converges put $f(x) = \sum_{n=0}^{\infty} C_n X^n \ (-1 < \times < 1)$ Then prove that $\lim_{x\to 1} f(x) = \sum_{n=0}^{\infty} C_n$.

Or

(b) Prove that every non constant polynomial with complex coefficients has a complex root.

14. (a) If E_1 and E_2 are measurable, then prove that $E_1 \cup E_2$ is measurable.

Or

(b) Let f and g be two measurable real–valued functions defined on the Some domain. Then prove that the function f_g is also measurable.

15. (a) If f and g are bounded measurable functions defined on a set E of finite measure, then prove that

(i) $\int_E (af + bg) = a \int_E f + b \int_E g$

(ii) If $f = g$ a.e. then $\int_E f = \int_E g$

Or

(b) State and Prove Fatou's lemma.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Suppose $f \in \mathcal{R}(\alpha)$ on $[a,b]$, $m \le f \le M, \phi$ is continuous on $[M,M]$ and $h(x) = \phi(f(x))$ on $[a,b]$. Then prove that $h \in \mathcal{R}(\alpha)$ on $[a,b]$.

17. State and prove the stone –Weierstrass theorem.

18. Derive striling's formula.

19. Prove that the outer measure of an interval is its length.

20. State and prove Lebeggul Convergence theorem.

AFS–0555

Sub. Code
7MMA2C3

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Mathematics

PARTIAL DIFFERENTIAL EQUATIONS

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Show that the direction Cosines of the tangent at the point (x, y, z) to the conic $ax^2 + by^2 + cz^2 = 1, x + y + z = 1$ are proportional to $(by - cz, cz - ax, ax - by)$.

2. Show that the equation $(x^2z - y^3)dx + 3xy^2dy + x^3dz = 0$ is integrable.

3. Eliminate the constants a and c from the equation $x^2 + y^2 + (z - c)^2 = a^2$.

4. Give an example of linear equation.

5. Find a complete integral of the equation $pq = 1$.

6. When we say that two surface are circumscribe to each other?

7. If $u = f(x+iy) + g(x-iy)$, where the functions f and g are arbitrary, then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

8. When we say that the equation $Rr + Ss + Tt + f(x, y, z, p, q) = 0$ is elliptic.

9. Write down the interior Dirichlet boundary value problem for Laplace's equations.

10. Write down the exterior Neumann boundary value problem.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Solve the equations $\frac{dx}{y+\alpha_z} = \frac{dy}{z-\beta x} - \frac{dz}{x+\gamma\, y}$.

Or

(b) Verify that the differential equation $(y^2 + yz)dx + (xz + z^2)\, dy + (y^2 - xy)dz = 0$ is integrable.

12. (a) Find the general solution of the differential equation $x^2 \frac{\partial z}{\partial x} + y^2 = \frac{\partial z}{\partial y} = (x+y)z$.

Or

(b) Find the surface which intersects the surfaces of the system $Z(x+y) = C(3z+1)$. orthogonally and which passes through the circle $x^2 + y^2 = 1, z = 1$.

13. (a) Find a complete integral of the equation $p^2x + q^2y = z$.

Or

(b) Find a complete integral of the equation $p^2y(1+x^2) = qx^2$..

14. (a) If $\alpha_r\ D + \beta_r D' + \gamma_r$ is a factor of $F(D, D')$ and $\phi_r(\varepsilon)$ is an arbitrary function of the single variable ε,, then prove that if $\alpha_r \neq 0,\ u_r = \exp\left(\frac{-\gamma_r x}{\alpha_r}\right)\phi_r(\beta_r\ x - \alpha_r y$ is a solution of the equation $F(D, D')Z = 0$.

Or

(b) Reduce the equation $\frac{\partial^2 z}{\partial x^2} = x^2 \frac{\partial^2 z}{\partial y^2}$ to Canonical form.

15. (a) Derive D'Alembert's solution of the ore–dimensional wave equation.

Or

(b) Prove that the solutions of a certain Neumann problem can differ from one another by a constant only.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Verify that the equation $z(z+y^2)dx + z(z+x^2)dy - xy(x+y)dz = 0$ is integrable and find its primitive.

17. Show that a necessary and sufficient condition that a surface be an integral surface of a partial differential equation is that at each point its tangent element should touch the elementary cone of the equation.

18. Show that the only integral surface of the equation $2q(z - px - qy) = 1 + q^2$ which is circumcribed about the poraboloid $2x = y^2 + z^2$ is the enveloping cylinder which touches it along its section by the plane $y + 1 = 0$.

19. Find the solution of the equation $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = x - y.$

20. Show that the Poisson integral $\theta(x,t) = \frac{1}{2(\pi kt)^{1/2}} \int_{-\infty}^{\infty} \phi(\varepsilon) \exp\left\{-\frac{(x-\varepsilon)^2}{4kt}\right\} d\varepsilon.$ is the solution of the initial value problem $\frac{\partial^2 \theta}{\partial x^2} = \frac{1}{k}\frac{\partial \theta}{\partial t}$ $-\infty < x < \infty, \theta(x,0) = \phi(x)$.

AFS–0556

Sub. Code
7MMA2C4

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Mathematics

MECHANICS

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. State the principle of virtual work.

2. Write down the Lagrangian form of d'Alembert's principle.

3. Define kinetic energy.

4. Define generalised inertial force.

5. State Hamilton's principle.

6. State Jacobi's theorem.

7. Write down the Jacobi's principle of least action.

8. Write down the modified Hamilton Jacobi's equation.

9. Define canonical transformation.

10. Define generation function.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain degrees of freedom.

Or

(b) Explain holonomic constraints.

12. (a) Suppose a mass spring system is attached to a frame which is translating with a uniform velocity v_0. Let l_0 be the unstressed spring length and use the elongation x as the generalised co-ordinate. Find the Jacobi integral for the system.

Or

(b) Explain Liouville's system.

13. (a) Find the stationary values of the function $f = x$, subject to the constraints $\phi_1 = x^2 + y^2 + z^2 - 4 = 0$; $\phi_2 = xy - 1 = 0$.

Or

(b) Explain modified Hamilton's principle.

14. (a) Explain conservative system and Ignorable co-ordinates.

Or

(b) Explain Hamilton's principle function.

15. (a) Consider the transformation $Q = \sqrt{e^{-2q} - p^2}$, $P = \cos^{-1}\left(pe^q\right)$. Use the poisson bracket to show that it is canonical.

Or

(b) Explain Lagrange brackets.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain virtual displacement.

17. Derive the standard form of Lagrange's equation for a holonomic system.

18. Explain Brachistochrone problem.

19. Explain pfaffian differential form.

20. Explain Bilinear covariant.

AFS–0557

Sub. Code
7MMA2E1

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Mathematics

Elective — GRAPH THEORY

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. If G is simple, then prove that $\varepsilon \leq \binom{\gamma}{2}$.

2. Define a connected graph. Give an example.

3. What is meant by a cut vertex? Give an example.

4. Draw the Petersen graph.

5. State the Berge theorem.

6. Write short notes on edge colouring.

7. Define an independent set of a graph. Give an example.

8. When will you say a graph is critical?

9. Define a planar graph with an illustration.

10. If G is a plane graph, then prove that $\sum_{f \in F} d(f) = 2\varepsilon$.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Define a spanning subgraph with an example. Also prove that $\delta \leq 2\varepsilon/\gamma \leq \Delta$.

Or

(b) If G us a tree, then prove that $\varepsilon = \gamma - 1$.

12. (a) With the usual notations, prove that $k \leq k' \leq \delta$.

Or

(b) If G is a simple graph with $\gamma \geq 3$ and $\delta \geq \gamma/2$, then prove that G is hamiltonian.

13. (a) If G is a K-regular bipartite graph with $k > 0$, then prove that G has a perfect matching.

Or

(b) If G is bipartite graph, then prove that $\chi' = \Delta$.

14. (a) What is meant by the edge independence number and edge covering number of G? Also prove that $\alpha' + \beta' = \gamma$ if $\delta > 0$.

Or

(b) Prove that every critical graph is a block.

15. (a) Show that k_5 is nonplanar.

Or

(b) State and prove Euler's formula for a connected plane graph.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. With the usual notations, prove that $\tau(K_n) = n^{n-2}$.

17. Prove that a nonempty connected graph is Eulerian if and only if it has no vertices of odd degree.

18. State and prove the Vizing's theorem.

19. With the usual notations, prove the following :

(a) $r(k, l) \leq \binom{k+l-2}{k-1}$;

(b) $r(k, k) \geq 2^{k/2}$.

20. State and prove the five colour theorem.

AFS–0558

Sub. Code
7MMA2E2

M.Sc. DEGREE EXAMINATION, APRIL 2018

Second Semester

Mathematics

Elective — APPLIED ALGEBRA

(CBCS – 2017 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. What is meant by covering machines?

2. When will you say states are compatible?

3. Write ALGOL expression of $\frac{6^{2.437}}{0.34^{3.26}}$.

4. Define the following terms:

   (a) Language;

   (b) Grammar

5. Write down the basic forms of conditional statements in ALGOL.

6. Define a Boolean subalgebra.

7. When will you say a Boolean function is symmetric?

8. Give an example of a flip-flop.

9. Define the following terms:

   (a) Encoding function;

   (b) Decoding function.

10. What is a group code?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Draw the state diagram for the following automation.

	γ		ξ	
	0	1	0	1
1	1	2	0	0
2	2	3	0	0
3	3	4	0	0
4	4	1	0	1

Or

   (b) If $G(E_r^1) - G(E_{r-1}^1) = \phi$, then prove that $G(E_r^1) = G(E_{r+k}^1)$ for all $k \geq 0$.

12. (a) Explain FOR statement in ALGOL. Write a program using FOR statement in ALGOL.

Or

   (b) Evaluate the f expressions for $A = 2$, $B = 3$, $C = -2$, $D = 4$, $E = 5$, $F = 6$ and $G = -2$.

   (i) $A \times D - E + F \uparrow G$;

   (ii) $A - B \times C + D$;

   (iii) $A \uparrow D - E \uparrow F$.

13. (a) State and prove the Dedekind lemma.

Or

(b) Find the disjunctive normal form for the following Boolean polynomials in x, y, z:

(i) y; (ii) $xy \vee z'$; (iii) $x + yz'$.

14. (a) State and prove the optimality principle.

Or

(b) Draw the following two-level gating networks for the proceeding f, using OR-to-AND.

15. (a) Enumerate a block code with an illustration.

Or

(b) What is a Hamming code? How is it formed?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Find the minimal machine equivalent to the following machine.

	γ		ξ	
	a	b	a	b
1	1	6	0	0
2	1	4	0	0
3	2	5	1	0
4	5	8	1	1
5	1	3	1	0
6	8	5	1	1
7	6	3	1	1
8	2	5	1	0

17. Write a program that multiples each $\overline{a_{ij}}$ in a matrix A of b.

18. In any Boolean algebra $\beta = [A, \wedge, \vee, ']$, prove that the relation $a \leqq b$ is partial ordering of A. Moreover, in terms of this partial ordering, prove that $a \wedge b = \text{glb}\{Q, b\}$ and $a \vee b = \text{lub}\{a, b\}$.

19. Explain the step-by-step procedure for finding the shortest path from node a to node b in any directed group $\overline{G}$ whose arcs α_i have specified positive lengths $\lambda(\alpha_i)$.

20. (a) Describe the systematic algebraic technique for encoding binary words by matrix multiplication.

    (b) Show that any group translation $x \mapsto x + c$ of the group $[z_2 r, +]$ preserves distance between code words.